

Harbour porpoise *Phocoena phocoena* click detectability estimates depend on behaviour and detection-classification method

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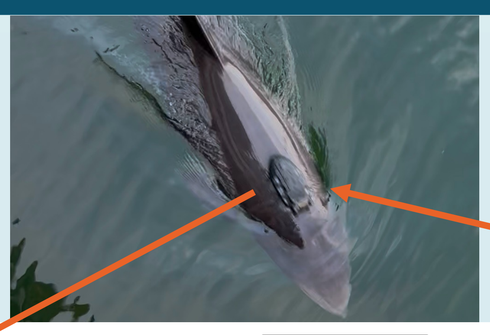
DTO-BioFlow: Integration of biodiversity monitoring data into the Digital Twin Ocean

Refs1-4 author affiliations

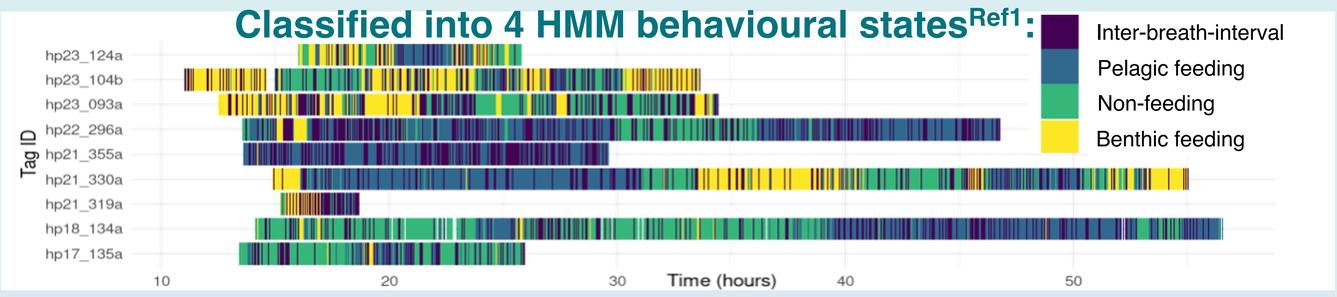
Echolocation click detection depends on: ***Animal movement*** *(due to narrow beam pattern) and* ***click rates***
During foraging, turning rates and click rates are increased

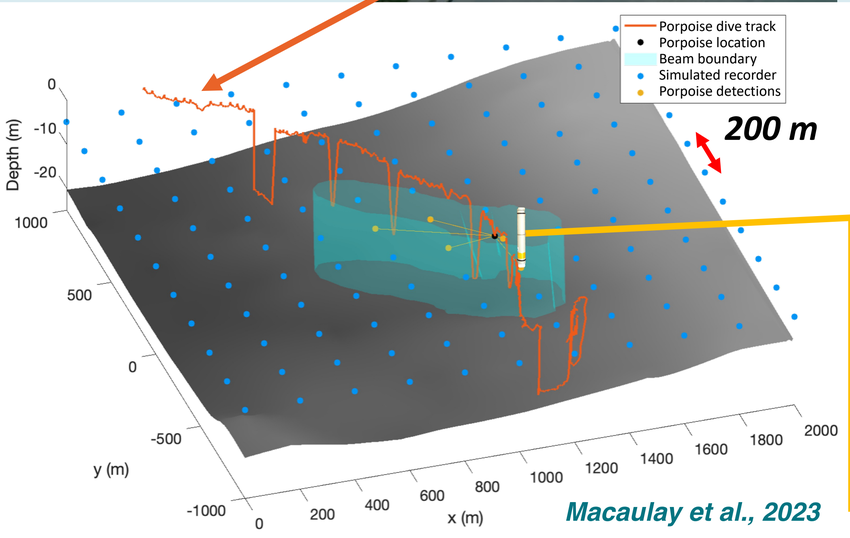
How do behaviours affect the probability of detecting clicks, and other commonly used summary detection units: ***Click sequences*** *(click 'trains' used for species identification) and* ***detection positive seconds/minutes****?*

Methods



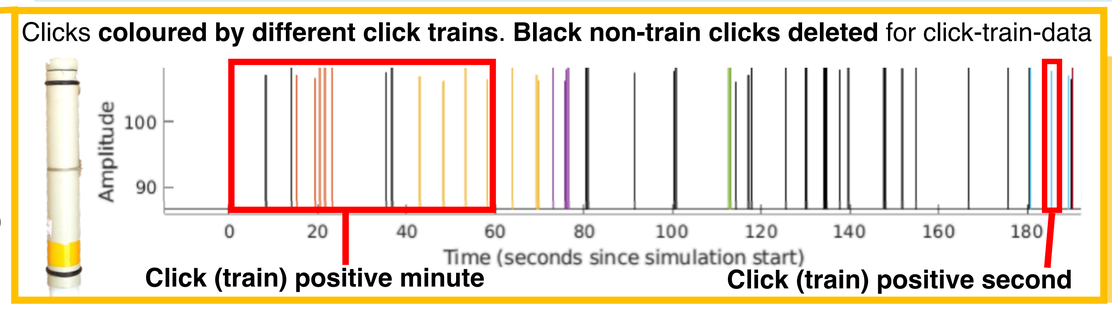
220hrs Harbour porpoise DTAG4 data





Macaulay et al., 2023

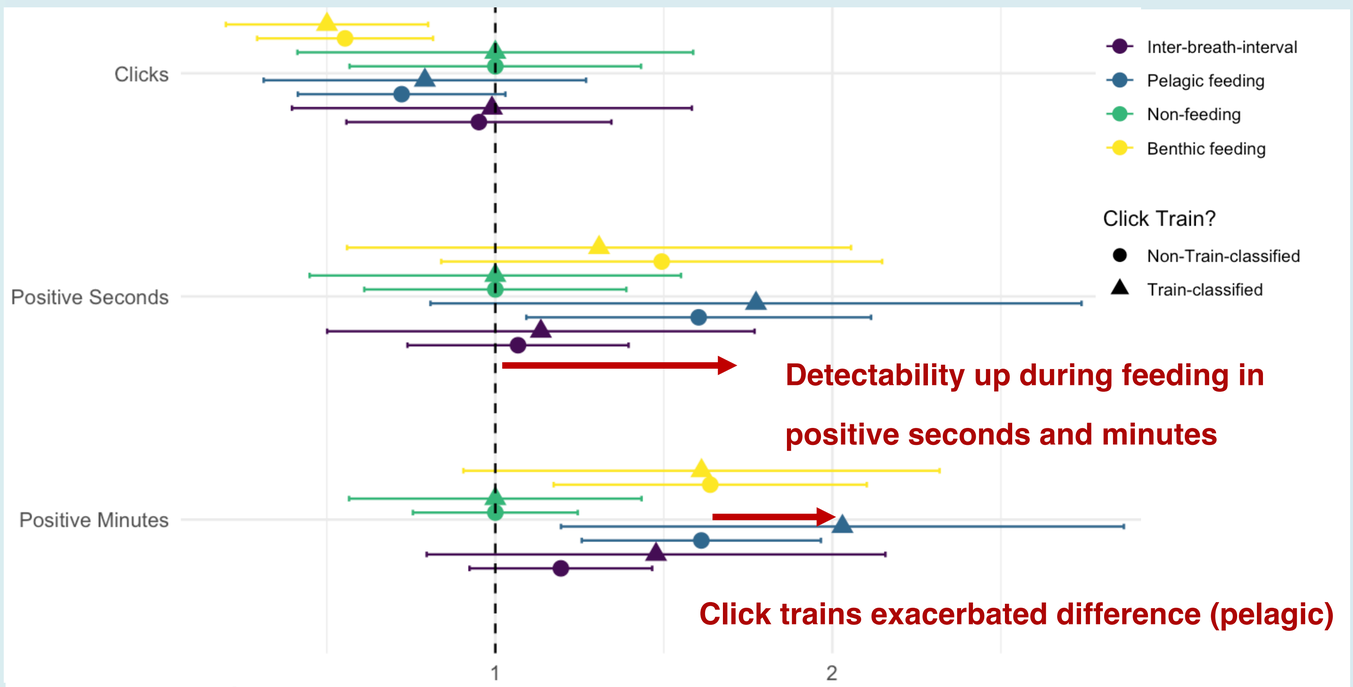
Simulate CPOD detectors around 3D tag track, producing **>100,000 detection time series like this:**



Custom **click train classifier** matching CPOD KERNO recall (~38%[Ref2, Ref3])

Results

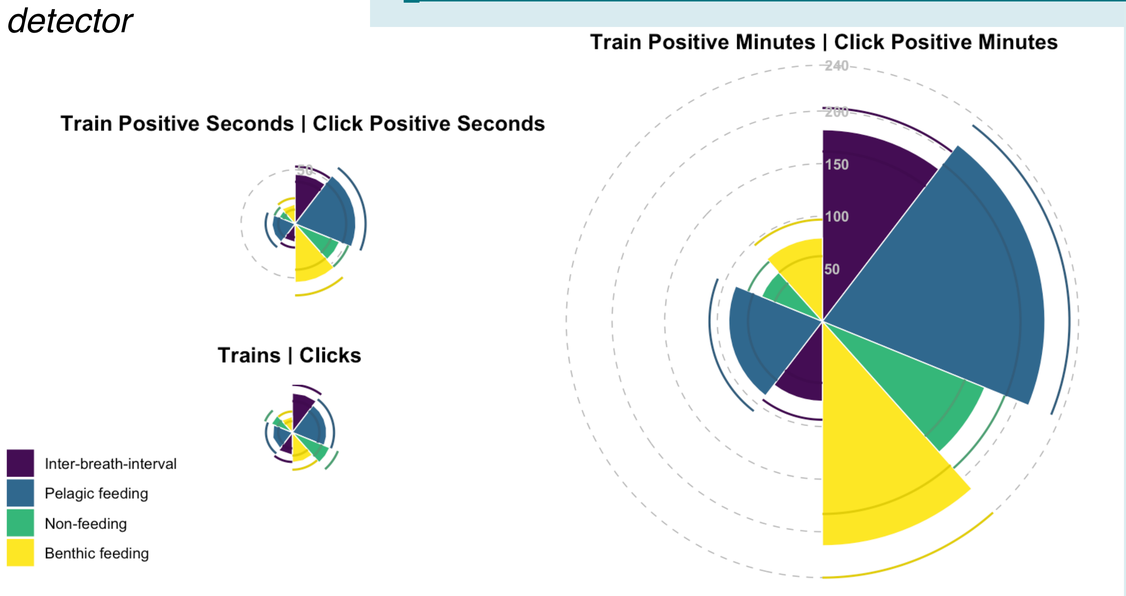
Detectability differences of behaviours



Predicted $\hat{P}$ *from GLMM as a ratio of 'Non-feeding' behavioural state (=1 in this plot) to illustrate differences in* $\hat{P}$ *between behaviours*

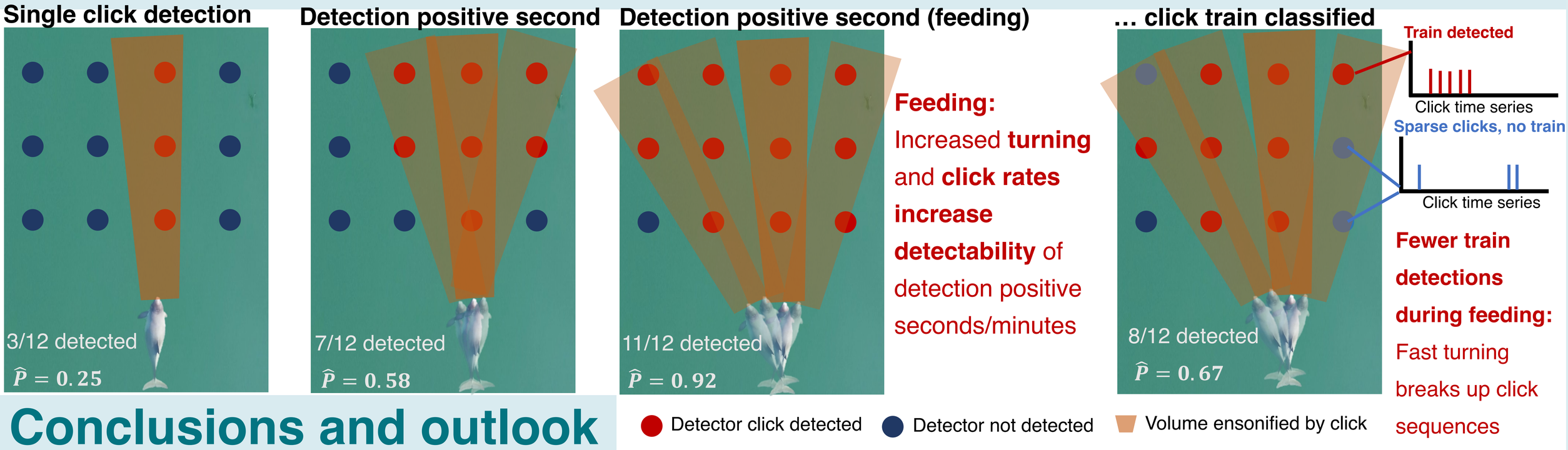
Effective detection area by behaviour

Expresses detectability as an area of certain detection around a detector



…6 detection units:

	Clicks	Click-Positive-Seconds	Click-Positive-Minutes
No train filter	All clicks	Seconds with at least 1 click	Minutes with at least 1 click
Click train filtered	All clicks in click trains	Seconds with at least 1 click in click trains	Minutes with at least 1 click in click trains



Conclusions and outlook

$\hat{P}$ **increases during feeding relative to non-feeding** in **positive seconds** and **minutes,** as reported previously[Ref4]

Implications: Detectability increases 1.5 - 2 fold during foraging in positive seconds & minutes. Problematic to transfer $\hat{P}$ between/within studies. Train classification can exacerbate this difference or alleviate it depending on behaviour. Improve **detection-classification robustness,** estimate **behavioural state from monitoring data** to account for $\hat{P}$